

Tutorial 2: Solutions

ES 336 – Computer Organization and Architecture

Indian Institute of Technology Gandhinagar

Problem 1:

A benchmark program is run on a 40 MHz processor. The executed program consists of 100,000 instruction executions, with the following instruction mix and clock cycle count:

Instruction Type	Instruction Count	Cycles per Instruction
Integer arithmetic	45000	1
Data transfer	32000	2
Floating point	15000	2
Control transfer	8000	2

Determine the effective CPI, MIPS rate, and execution time for this program.

Solution 1:

Given:

- Clock frequency = 40 MHz
- Clock cycle time = $\frac{1}{40 \times 10^6} = 25 \text{ ns}$
- Total instructions = 100,000

Step 1: Calculate total clock cycles

$$\text{Total clock cycles} = (45000 \times 1) + (32000 \times 2) + (15000 \times 2) + (8000 \times 2)$$

$$\text{Total clock cycles} = 45000 + 64000 + 30000 + 16000 = 155000 \text{ cycles}$$

Step 2: Calculate effective CPI

$$\text{CPI} = \frac{\text{Total Clock Cycles}}{\text{Total Instructions}} = \frac{155000}{100000} = 1.55$$

Step 3: Calculate execution time

$$\text{Execution Time} = \text{Total Clock Cycles} \times \text{Clock Cycle Time}$$

$$= 155000 \times 25 \times 10^{-9} = 3.875 \times 10^{-3} \text{ seconds} = 3.875 \text{ ms}$$

Step 4: Calculate MIPS rate

$$\text{MIPS} = \frac{\text{Instruction Count}}{\text{Execution Time} \times 10^6} = \frac{100000}{3.875 \times 10^{-3} \times 10^6} = \frac{100000}{3875} = 25.81 \text{ MIPS}$$

$$\text{Alternatively: MIPS} = \frac{\text{Clock Rate}}{\text{CPI} \times 10^6} = \frac{40 \times 10^6}{1.55 \times 10^6} = 25.81 \text{ MIPS}$$

Answer: CPI = 1.55, MIPS = 25.81, Execution Time = 3.875 ms

Problem 2:

Our favorite program runs in 10 seconds on computer A, which has a 2 GHz clock. We are trying to help a computer designer build a computer, B, which will run this program in 6 seconds. The designer has determined that a substantial increase in the clock rate is possible, but this increase will affect the rest of the CPU design, causing computer B to require 1.2 times as many clock cycles as computer A for this program. What clock rate should we tell the designer to target?

Solution 2:

Given:

- Computer A: Clock rate = 2 GHz, Execution time = 10 seconds
- Computer B: Target execution time = 6 seconds, Clock cycles = 1.2 times Computer A

Step 1: Find clock cycles for Computer A

$$\text{Clock Cycles}_A = \text{Clock Rate}_A \times \text{Execution Time}_A = 2 \times 10^9 \times 10 = 2 \times 10^{10} \text{ cycles}$$

Step 2: Find clock cycles for Computer B

$$\text{Clock Cycles}_B = 1.2 \times \text{Clock Cycles}_A = 1.2 \times 2 \times 10^{10} = 2.4 \times 10^{10} \text{ cycles}$$

Step 3: Calculate required clock rate for Computer B

$$\text{Clock Rate}_B = \frac{\text{Clock Cycles}_B}{\text{Execution Time}_B} = \frac{2.4 \times 10^{10}}{6} = 4 \times 10^9 \text{ Hz} = 4 \text{ GHz}$$

Answer: The designer should target a clock rate of 4 GHz.

Problem 3:

Early examples of CISC and RISC design are the VAX 11/780 and the IBM RS/6000, respectively. Using a typical benchmark program, the following machine characteristics

result:

Processor	Clock Frequency	Performance	CPU Time
VAX 11/780	5 MHz	1 MIPS	$12x$ seconds
IBM RS/6000	25 MHz	18 MIPS	x seconds

The final column shows that the VAX required 12 times longer than the IBM measured in CPU time.

- What is the relative size of the instruction count of the machine code for this benchmark program running on the two machines?
- What are the *CPI* values for the two machines?

Solution 3:

Given:

- VAX 11/780: 5 MHz, 1 MIPS, CPU time = $12x$ seconds
- IBM RS/6000: 25 MHz, 18 MIPS, CPU time = x seconds

Part (a): Relative instruction count

Using: $\text{MIPS} = \frac{\text{Instruction Count}}{\text{CPU Time} \times 10^6}$

For VAX: $\text{IC}_{\text{VAX}} = \text{MIPS} \times \text{CPU Time} \times 10^6 = 1 \times 12x \times 10^6 = 12x \times 10^6$

For IBM: $\text{IC}_{\text{IBM}} = 18 \times x \times 10^6 = 18x \times 10^6$

Ratio: $\frac{\text{IC}_{\text{VAX}}}{\text{IC}_{\text{IBM}}} = \frac{12x \times 10^6}{18x \times 10^6} = \frac{12}{18} = \frac{2}{3}$

Answer (a): The VAX instruction count is $\frac{2}{3}$ (or 0.667) times that of the IBM RS/6000.

Part (b): CPI values

Using: $\text{CPI} = \frac{\text{Clock Rate}}{\text{MIPS} \times 10^6}$

For VAX: $\text{CPI}_{\text{VAX}} = \frac{5 \times 10^6}{1 \times 10^6} = 5$

For IBM: $\text{CPI}_{\text{IBM}} = \frac{25 \times 10^6}{18 \times 10^6} = 1.39$

Answer (b): CPI for VAX = 5, CPI for IBM RS/6000 = 1.39

Problem 4:

While browsing at Billy Bob's computer store, you overhear a customer asking Billy Bob what the fastest computer in the store is that he can buy. Billy Bob replies, "You're looking at our Macintoshes. The fastest Mac we have runs at a clock speed of 1.2 gigahertz. If you really want the fastest machine, you should buy our 2.4-gigahertz Intel Pentium IV instead." Is Billy Bob correct? What would you say to help this customer?

Solution 4:

Billy Bob is not necessarily correct. Clock speed alone does not determine the fastest computer. The actual performance depends on multiple factors:

- **CPI (Cycles Per Instruction):** Different architectures execute instructions with different efficiencies
- **Instruction count:** The same program may require different numbers of instructions on different architectures
- **Architecture and optimization:** Compiler quality, cache design, pipeline efficiency, and other microarchitectural features significantly impact performance

Performance should be measured by execution time for actual workloads, not just clock speed. The customer should ask:

1. What tasks will you primarily be running?
2. Are there benchmarks comparing these systems on similar workloads?
3. What is the actual performance (instructions per second, or better yet, real application performance) rather than just clock rate?

The 1.2 GHz Mac could potentially outperform the 2.4 GHz Pentium IV for certain tasks if it has better IPC (instructions per cycle) or architectural advantages.

Problem 5:

Some microprocessors today are designed to have adjustable voltage, so a 15% reduction in voltage may result in a 15% reduction in frequency. What would be the impact on dynamic energy and dynamic power?

Solution 5:

Given: 15% reduction in voltage results in 15% reduction in frequency.

Let initial voltage = V , new voltage = $0.85V$

Let initial frequency = f , new frequency = $0.85f$

Dynamic Energy per operation:

$$E_{\text{dynamic}} \propto CV^2$$

New energy: $E_{\text{new}} = C(0.85V)^2 = C \times 0.7225V^2 = 0.7225E_{\text{old}}$

Reduction in energy: $(1 - 0.7225) \times 100\% = 27.75\%$ reduction

Dynamic Power:

$$P_{\text{dynamic}} \propto CV^2f$$

New power: $P_{\text{new}} = C(0.85V)^2(0.85f) = C \times 0.7225 \times 0.85 \times V^2f = 0.6141P_{\text{old}}$

Reduction in power: $(1 - 0.6141) \times 100\% = 38.59\%$ reduction

Answer: Dynamic energy reduces by 27.75%, dynamic power reduces by 38.59%.

Problem 6:

Server farms such as Google and Yahoo! provide enough compute capacity for the highest request rate of the day. Imagine that most of the time these servers operate at only 60% capacity. Assume further that the power does not scale linearly with the load; that is, when the servers are operating at 60% capacity, they consume 90% of maximum power. The servers could be turned off, but they would take too long to restart in response to more load. A new system has been proposed that allows for a quick restart but requires 20% of the maximum power while in this "barely alive" state.

1. How much power savings would be achieved by turning off 60% of the servers?
2. How much power savings would be achieved by placing 60% of the servers in the "barely alive" state?
3. How much power savings would be achieved by reducing the voltage by 20% and frequency by 40%?
4. How much power savings would be achieved by placing 30% of the servers in the "barely alive" state and 30% off?

Solution 6:**Given assumptions:**

- Normal operation: 60% capacity consumes 90% maximum power
- Barely alive state: consumes 20% maximum power
- Off state: consumes 0% power (but slow restart)

Let maximum power = P_{max}

Current power consumption with all servers at 60% capacity = $0.9P_{\text{max}}$

Part (a): Turning off 60% of servers

40% servers at 60% capacity: $0.4 \times 0.9P_{\text{max}} = 0.36P_{\text{max}}$

60% servers off: $0.6 \times 0 = 0$

$$\text{Total power} = 0.36P_{\max}$$

$$\text{Power savings} = 0.9P_{\max} - 0.36P_{\max} = 0.54P_{\max} \text{ or } \mathbf{60\% \text{ savings}}$$

Part (b): Placing 60% of servers in "barely alive" state

$$40\% \text{ servers at } 60\% \text{ capacity: } 0.4 \times 0.9P_{\max} = 0.36P_{\max}$$

$$60\% \text{ servers barely alive: } 0.6 \times 0.2P_{\max} = 0.12P_{\max}$$

$$\text{Total power} = 0.48P_{\max}$$

$$\text{Power savings} = 0.9P_{\max} - 0.48P_{\max} = 0.42P_{\max} \text{ or } \mathbf{46.67\% \text{ savings}}$$

Part (c): Reducing voltage by 20% and frequency by 40%

$$\text{New voltage} = 0.8V, \text{ new frequency} = 0.6f$$

$$\text{Power} \propto V^2f, \text{ so new power} = (0.8)^2 \times 0.6 = 0.64 \times 0.6 = 0.384 \text{ of original}$$

$$\text{All servers at reduced power: } 0.384 \times 0.9P_{\max} = 0.3456P_{\max}$$

$$\text{Power savings} = 0.9P_{\max} - 0.3456P_{\max} = 0.5544P_{\max} \text{ or } \mathbf{61.6\% \text{ savings}}$$

Part (d): 30% barely alive and 30% off

$$40\% \text{ servers at } 60\% \text{ capacity: } 0.4 \times 0.9P_{\max} = 0.36P_{\max}$$

$$30\% \text{ servers barely alive: } 0.3 \times 0.2P_{\max} = 0.06P_{\max}$$

$$30\% \text{ servers off: } 0.3 \times 0 = 0$$

$$\text{Total power} = 0.42P_{\max}$$

$$\text{Power savings} = 0.9P_{\max} - 0.42P_{\max} = 0.48P_{\max} \text{ or } \mathbf{53.33\% \text{ savings}}$$

Problem 7:

Find the number of dies per 300 mm (30 cm) wafer for a die that is 1.5 cm on a side and for a die that is 1.0 cm on a side.

Solution 7:

Formula for dies per wafer:

$$\text{Dies per wafer} = \frac{\text{Wafer area}}{\text{Die area}} - \frac{\pi \times \text{Wafer diameter}}{\sqrt{2} \times \text{Die area}}$$

Given: Wafer diameter = 300 mm = 30 cm, so radius = 15 cm

$$\text{Wafer area} = \pi r^2 = \pi \times 15^2 = 706.86 \text{ cm}^2$$

Case 1: Die size = 1.5 cm × 1.5 cm

$$\text{Die area} = 1.5^2 = 2.25 \text{ cm}^2$$

$$\text{Dies per wafer} = \frac{706.86}{2.25} - \frac{\pi \times 30}{\sqrt{2} \times 2.25} = 314.16 - \frac{94.25}{2.12} = 314.16 - 44.46 = 269.7 \approx 270 \text{ dies}$$

Case 2: Die size = 1.0 cm × 1.0 cm

$$\text{Die area} = 1.0^2 = 1.0 \text{ cm}^2$$

$$\text{Dies per wafer} = \frac{706.86}{1.0} - \frac{\pi \times 30}{\sqrt{2} \times 1.0} = 706.86 - \frac{94.25}{1.414} = 706.86 - 66.65 = 640.21 \approx 640 \text{ dies}$$

Answer: 270 dies for 1.5 cm die, 640 dies for 1.0 cm die.

Problem 8:

Find the die yield for dies that are 1.5 cm on a side and 1.0 cm on a side, assuming a defect density of 0.031 per cm², and N is 13.5.

Solution 8:

Die yield formula (Bose-Einstein):

$$\text{Yield} = \left(1 + \frac{\text{Defects per unit area} \times \text{Die area}}{N} \right)^{-N}$$

Given: Defect density = 0.031 per cm², $N = 13.5$

Case 1: Die area = 2.25 cm² (1.5 cm × 1.5 cm)

$$\begin{aligned} \text{Yield} &= \left(1 + \frac{0.031 \times 2.25}{13.5} \right)^{-13.5} = \left(1 + \frac{0.06975}{13.5} \right)^{-13.5} \\ &= (1 + 0.005167)^{-13.5} = (1.005167)^{-13.5} = 0.9323 = 93.23\% \end{aligned}$$

Case 2: Die area = 1.0 cm² (1.0 cm × 1.0 cm)

$$\begin{aligned} \text{Yield} &= \left(1 + \frac{0.031 \times 1.0}{13.5} \right)^{-13.5} = \left(1 + \frac{0.031}{13.5} \right)^{-13.5} \\ &= (1 + 0.002296)^{-13.5} = (1.002296)^{-13.5} = 0.9695 = 96.95\% \end{aligned}$$

Answer: Yield for 1.5 cm die = 93.23%, yield for 1.0 cm die = 96.95%.

Problem 9:

You are designing a system for a real-time application that requires meeting specific deadlines. Finishing the computation faster gains nothing. You find that your system can execute the necessary code, in the worst case, twice as fast as necessary.

- a. How much energy do you save if you execute at the current speed and turn off the system when the computation is complete?

- b. How much energy do you save if you set the voltage and frequency to be half as much?

Solution 9:

Given: System can execute twice as fast as necessary in worst case.

Part (a): Execute at current speed and turn off

Let the required time be T and energy at full speed be E .

Power $P = \frac{E}{T}$, so $E = P \times T$

Since system runs twice as fast, it completes in time $\frac{T}{2}$ at full power, then is off for time $\frac{T}{2}$.

Energy consumed = $P \times \frac{T}{2} = \frac{E}{2}$

Energy savings = 50%

Part (b): Set voltage and frequency to half

When frequency is halved, the execution time doubles (back to the required time T).

When voltage is halved: $V_{\text{new}} = 0.5V$

Power $\propto V^2 f$

New power = $(0.5)^2 \times 0.5 \times P = 0.25 \times 0.5 \times P = 0.125P$

Energy consumed = $0.125P \times T = 0.125E$

Energy savings = 87.5%

Answer: (a) 50% energy savings, (b) 87.5% energy savings. Part (b) is more energy-efficient.